Problem Set 2

Before you begin, please read the following instructions carefully:

- Start early.
- Your submission should include:
  - One document with the answers
  - The code used to answer the questions that require the `R`.
  - If you use R markdown, then submit `*.Rnw` file plus the compiled document.
- Late submission will not be accepted. Submission date: before Lab on Oct 15th.
- For the questions which require the use of `R`, make sure you include fully annotated code as part of your answers. For mathematical questions, you should include your detailed derivations.

Questions:

1. Imagine that $(y_1, y_2, y_3, y_4, y_5) \sim \text{Mult}(n, \delta_1, \delta_2, \delta_3, \delta_4, \delta_5)$, with $\delta_1 = \frac{1}{4}\theta$, $\delta_2 = \delta_3 = \frac{1}{4}(1-\theta)$, $\delta_4 = \frac{1}{2}$, and $\delta_5 = \frac{1}{4}\theta$. Instead of observing these data, we observe $(x_1, x_2, x_3, x_4) \sim \text{Mult}(n, \pi_1, \pi_2, \pi_3, \pi_4)$, where $\pi_1 = \frac{1}{4}\theta$, $\pi_2 = \pi_3 = \frac{1}{4}(1-\theta)$, and $\pi_4 = \frac{1}{2} + \frac{1}{4}\theta$. So we have that $x_4 = y_4 + y_5$. Use the EM algorithm to recover an estimate of θ.

2. Let M_{ij} indicate whether records i and j are a match. We do not observe M_{ij}, but we do observe their agreement on K variables. Define $\delta_k(i,j) = 1$ if the records agree on variable k, and 0 otherwise. Each pair has an observed comparison vector

$$\boldsymbol{\delta}(i,j) = \big(\delta_1(i,j), \ldots, \delta_K(i,j)\big)$$

   For n record pairs, use the EM algorithm to estimate the probability that each pair is a match, along with the model parameters. Assume that conditional on M_{ij} each component of $\boldsymbol{\delta}(i,j)$ is independent of each other.

   Hint: given the information above, if you make the right calls for the distributions of M_{ij} and of $\boldsymbol{\delta}(i,j) \mid M_{ij} = m$, where $m \in \{0,1\}$, then you will be good to go.

3. Consider the following table of counts for the agreement patterns $\boldsymbol{\delta}$ defined in the previous question:

δ_1	δ_2	δ_3	Count
0	0	0	172738
1	0	0	2030
0	1	0	114
0	0	1	68
1	1	1	50

   Here, $\boldsymbol{\delta} = (0,0,0)$ indicates disagreement on all four fields and occurs for 172738 record pairs. Similarly, $\boldsymbol{\delta} = (1,1,1)$ indicates agreement on all four fields and occurs for 50 record pairs. Patterns omitted from the table have a count of zero.

   These counts are the observed data; as in the previous question, the match status M remains unobserved for every pair. Using your derivations from the previous question, implement the EM algorithm and fit the model to these counts.

Store the observed-data log-likelihood at each iteration. Plot it against the iteration number and verify that it does not decrease.

4. Suppose X_i with $i = \{1, \ldots, n\}$, comes from a two-component Gaussian mixture. For n_1 observations, the class label Z_i is known; for the rest i.e., $n_2 = n - n_1$, the class labels are missing. Derive the EM algorithm for this problem.